

Subject : Burgers equation
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1 Burgers equation

The Burgers equation reads (Debnath (2008)):

$$\boxed{\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0, \quad u > 0} \quad (1)$$

Because we will use a finite volume method to solve this equation, the conservative form will be used, i.e.:

$$\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x} \left(\nu \frac{\partial u}{\partial x} \right) = 0 \quad (2)$$

Applying the finite volume approach:

$$\int_{\Omega} \frac{\partial u}{\partial t} d\Omega + \frac{1}{2} \int_{\Omega} \frac{\partial u^2}{\partial x} d\Omega - \int_{\Omega} \frac{\partial}{\partial x} \left(\nu \frac{\partial u}{\partial x} \right) d\Omega = 0 \quad (3)$$

and applying Green's theorem yields:

$$\int_{\Omega} \frac{\partial u}{\partial t} d\Omega + \frac{1}{2} \oint_{\Gamma} u^2 d\Gamma - \oint_{\Gamma} \nu \frac{\partial u}{\partial x} d\Gamma = 0 \quad (4)$$

2 Discretization in time and space

The discretization in space and the discretization in space are discussed in separate sections. Due to the applying regularization procedure the viscosity (ν) is increased by an artificial viscosity (Ψ), so the the implementation of $\nu + \Psi$ is a function of x .

2.1 Discretization in time

The used discretization in time reads:

$$\begin{aligned} \int_{\Omega} \frac{\partial u}{\partial t} d\Omega \approx \Delta x \Delta t_{inv} & \left(\frac{1}{8} \Delta u_{i-1}^{n+1,p+1} + \frac{6}{8} \Delta u_i^{n+1,p+1} + \frac{1}{8} \Delta u_{i+1}^{n+1,p+1} \right) + \\ & + \Delta x \Delta t_{inv} \left(\frac{1}{8} (u_{i-1}^{n+1,p} - u_{i-1}^n) + \frac{6}{8} (u_i^{n+1,p} - u_i^n) + \frac{1}{8} (u_{i+1}^{n+1,p} - u_{i+1}^n) \right) \end{aligned} \quad (5)$$

2.2 Discretization in place

Convection-term

$$\frac{1}{2} \oint_{\Gamma} u^2 d\Gamma = \frac{1}{2} u^2|_{i+\frac{1}{2}} - \frac{1}{2} u^2|_{i-\frac{1}{2}} \quad (6)$$

with the approximation $u^2 = (u^{n+\theta,p})^2 + 2u^{n+\theta,p}\theta\Delta u$ the discretization reads:

$$\approx u_{i+\frac{1}{2}}^{n+\theta,p}\theta\Delta u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}}^{n+\theta,p}\theta\Delta u_{i-\frac{1}{2}} + \frac{1}{2} \left(u_{i+\frac{1}{2}}^{n+\theta,p}\right)^2 - \frac{1}{2} \left(u_{i-\frac{1}{2}}^{n+\theta,p}\right)^2 \quad (7)$$

Viscosity term

With $\varepsilon = \nu + \Psi$ the viscosity term reads:

$$-\oint_{\Gamma} \varepsilon \frac{\partial u}{\partial x} d\Gamma = \left(\varepsilon \frac{\partial u}{\partial x}\right)_{i+\frac{1}{2}} - \left(\varepsilon \frac{\partial u}{\partial x}\right)_{i-\frac{1}{2}} \quad (8)$$

approximated by

$$\Rightarrow \theta \frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x} \Delta u_{i+1}^{n+1,p+1} - \theta \left(\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x} + \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x} \right) \Delta u_i^{n+1,p+1} - \theta \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x} \Delta u_{i-1}^{n+1,p+1} + \quad (9)$$

$$+ \left(\varepsilon_{i+\frac{1}{2}} \frac{u_{i+1}^{n+\theta,p} - u_i^{n+\theta,p}}{\Delta x} - \varepsilon_{i-\frac{1}{2}} \frac{u_i^{n+\theta,p} - u_{i-1}^{n+\theta,p}}{\Delta x} \right) \quad (10)$$

3 Energy Equation of the Inviscid Burgers' Equation

The **inviscid** Burgers' equation is given by:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0. \quad (11)$$

The kinetic energy per unit mass is:

$$E = \frac{1}{2} u^2 \quad (12)$$

The time derivative of the energy reads:

$$\frac{\partial E}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} u^2 \right) = u \frac{\partial u}{\partial t} = u \left(-u \frac{\partial u}{\partial x} \right) = -u^2 \frac{\partial u}{\partial x} = -\frac{\partial}{\partial x} \left(\frac{1}{3} u^3 \right) \quad (13)$$

So the energy equation becomes:

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{3} u^3 \right) = 0 \quad (14)$$

4 Boundary

Because it is a one dimensional equation we have only boundary conditions at the west- and east-side of the domain.

4.1 West boundary

This is an inflow boundary ($u > 0$).

4.2 East boundary

This is an outflow boundary ($u > 0$).

5 Numerical experiment

5.1 Standard Burgers equation

5.2 With source term

This experiment is taken from [Colombo et al. \(2026\)](#) and read

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = u^2, \quad u > 0 \quad (15)$$

With initial condition

$$u(x, 0) = \exp x + 0.3 \exp(-200(x + 0.5)^2) \quad (16)$$

and boundary condition at the left side

$$u(x_a, t) = \exp(x_a) \quad (17)$$

References

Colombo, C., C. Dalmaso, L. O. Müller, and A. Siviglia (2026). “Well-balanced high-order method for non-conservative hyperbolic PDEs with source terms: Application to one-dimensional blood flow equations with gravity”. In: *Journal of Computation Physics*. File: Colombo_Application to one-dimensional blood_2026.pdf.

Debnath, Lokenath (2008). *Sir Lighthill and Modern Fluid Dynamics*. Imperial College Press.